

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

Machine Learning Engineer with around 1.5 years of experience, currently pursuing an M.Tech in AI and ML at BITS Pilani. As a Research Associate, AI at Keywords Studios, I work with LLMs, Agentic AI, and full ML pipelines to build production systems that hold up in the real world, and was named "Rookie of the Year 2025" for beating every performance KPI. I love turning research into reliable systems people can count on. Outside work, I enjoy drawing, tennis, running, philosophy, ASL, reading, and exploring new technology.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani)	July 2026 – July 2028
M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing) <i>Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning</i>	
K.R. Mangalam University	July 2021 – July 2025
B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon	

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana	May 2025 – Present
- Awarded "Rookie of the Year 2025" for surpassing all performance KPIs through advanced model training, data-pipeline optimization, and cross-functional collaboration. - Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability. - Contributed to a cutting-edge web-automation AI agent that visually interprets on-screen content and, from a natural-language instruction, autonomously plans and executes browser actions such as clicking, typing, and navigating to complete complex, multi-step tasks. - Trained, evaluated, and quality-controlled the agent with a global team, curating high-quality datasets and running structured test reviews to deliver client-ready iterations with improved action accuracy and reliability. - Built custom data pipelines and delivered 85% of self-generated training datasets directly to production as the first member of a global team, accelerating model readiness.	
Artificial Intelligence Intern, Infosys – Remote	November 2024 – February 2025
- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and elevating customer-AI interaction quality. - Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process. - Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing hallucinations by 20% and making customer conversations feel more natural and trustworthy. - Collaborated with a team of 12 engineers to integrate end-to-end ML workflows and earned selection to present at the Best Project by Indian Interns initiative.	

PROJECTS

Dynamic Screen Companion	June 2025 – Present
- Built a real-time multi-modal assistant integrating the Gemini API to analyze live screen activity and deliver continuous voice feedback and interactive support. - Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by 40%. - Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain 95%+ extraction accuracy across varied screen layouts.	
Diabetic Optiscan	January 2025 – June 2025
- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a peer-reviewed research paper explaining how the system was designed and the accuracy improvements it achieved. - Led integration of a Vision Transformer (ViT-B/16) model on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in high-resolution retinal images. - Designed a custom wavelet modification technique that boosted detection precision by 50% over legacy methods.	
Currency Detection App for Visually Impaired	July 2024 – December 2024
- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud. - Conducted user interviews with visually impaired individuals to refine real-world accessibility features.	

SKILLS

Languages & Frameworks: Python, C++, SQL, R, FastAPI, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph
AI & Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Deep Learning, Transformers, Hugging Face, BERT, NLP, Prompt Engineering, Model Evaluation & Optimization, OOP, DSA
MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Model Serving, Model Monitoring, Data Pipelines, Git, Vector Databases (FAISS/Chroma)
Data & Cloud Tools: AWS, GCP (Vertex AI), Power BI, MySQL, MongoDB, Pandas, NumPy, Statistical Analysis

CERTIFICATIONS

Ducat India – Data Science Professional Training	April 2024 – April 2025
<i>Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.</i>	
AI/ML for Geodata Analysis – ISRO	
<i>Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.</i>	